

Reduced Group C^* -Algebras and the Ball-Covering Property

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1. *Let G be a discrete group. Then*

$$C_r^*(G) \text{ has BCP} \iff C_r^*(G) \text{ has UBCP} \iff G \text{ is countable.}$$

Idea of the Proof

The positive direction is separability. If G is countable, then the group algebra with rational complex coefficients is countable and dense in $C_r^*(G)$, so $C_r^*(G)$ has UBCP.

For the negative direction, countably many centres cannot see an uncountable group. Each centre belongs to the reduced algebra of a countable subgroup. After collecting countably many centres, choose one countable subgroup $H \leq G$ that contains all of them. Pick $g \in G \setminus H$. The canonical conditional expectation

$$E_H : C_r^*(G) \rightarrow C_r^*(H)$$

fixes the centres and kills λ_g . Hence the unitary λ_g is at distance at least $\|a_n\|$ from every centre a_n , so it escapes all BCP balls.

Proof

We use the standard equivalent BCP test: a normed space X has BCP if and only if there are countably many centres $x_n \in X$ such that

$$S_X \subseteq \bigcup_n B(x_n, \|x_n\|).$$

Lemma 1 (countable support subgroups). *For every $a \in C_r^*(G)$, there is a countable subgroup $H \leq G$ such that*

$$a \in C_r^*(H) \subseteq C_r^*(G).$$

Consequently, every countable subset of $C_r^(G)$ is contained in $C_r^*(H)$ for some countable subgroup $H \leq G$.*

Proof. The algebraic group algebra $\mathbb{C}[G]$ is dense in $C_r^*(G)$. Choose a sequence $f_m \in \mathbb{C}[G]$ such that $f_m \rightarrow a$ in norm. Let

$$S = \bigcup_m \text{supp}(f_m),$$

a countable subset of G , and let $H = \langle S \rangle$ be the subgroup generated by S . Then H is countable, each f_m belongs to $\mathbb{C}[H]$, and therefore $a \in C_r^*(H)$ under the canonical inclusion $C_r^*(H) \subseteq C_r^*(G)$.

For countably many elements, take the subgroup generated by the union of the countably many subgroups just obtained. $\square$

Lemma 2 (subgroup expectation). *If $H \leq G$, there is a contractive conditional expectation*

$$E_H : C_r^*(G) \rightarrow C_r^*(H)$$

such that

$$E_H(\lambda_s) = \begin{cases} \lambda_s, & s \in H, \\ 0, & s \notin H, \end{cases}$$

for the left regular unitaries λ_s .

Proof. On $\ell^2(G)$, identify $\ell^2(H)$ with the closed span of $\{\delta_h : h \in H\}$. The Fourier-coefficient map displayed above is the standard reduced subgroup expectation. It is completely positive and contractive because it is implemented by compression onto the coset decomposition of $\ell^2(G)$, or equivalently by the canonical coefficient-preserving expectation on the reduced crossed-section completion. On the dense algebra $\mathbb{C}[G]$ it has exactly the stated formula, so it extends to $C_r^*(G)$. $\square$

Proof of Theorem 1. If G is countable, then $C_r^*(G)$ is separable. Every separable normed space has UBCP: choose a countable dense subset (d_j) of the unit sphere and use the balls $B(2d_j, 3/2)$. Thus $C_r^*(G)$ has UBCP, hence BCP.

Assume now that G is uncountable, and let (a_n) be any countable family in $C_r^*(G)$. By Lemma 1, there is a countable subgroup $H \leq G$ such that $a_n \in C_r^*(H)$ for every n . Choose $g \in G \setminus H$, and put

$$u = \lambda_g \in C_r^*(G).$$

Then u is unitary, so $\|u\| = 1$. By Lemma 2,

$$E_H(u) = 0, \quad E_H(a_n) = a_n.$$

Therefore, for every n ,

$$\|u - a_n\| \geq \|E_H(u - a_n)\| = \| -a_n\| = \|a_n\|.$$

Thus $u \in S_{C_r^*(G)}$ lies outside every open ball

$$B(a_n, \|a_n\|).$$

Since the countable family of centres was arbitrary, $C_r^*(G)$ fails BCP. □

Scope and Limitations

This theorem gives a complete answer for reduced group C^* -algebras of discrete groups. It is a non-Type-I visibility theorem: the bad direction is detected by a canonical conditional expectation onto the countable subgroup generated by the proposed centres.

For uncountable groups whose reduced C^* -algebras are simple, this gives simple one-point-primitive non-Type-I C^* -algebras without BCP, independent of the von Neumann factor obstruction. The result does not classify arbitrary non-Type-I C^* -algebras; it identifies one robust mechanism by which nonseparability is visible in the ambient norm.

Novelty and Search Bounds

The local indexes were searched for arXiv:2311.14257, reduced group C^* -algebras, non-Type-I, conditional expectations, diagonal escape, BCP, and UBCP. Web searches on June 17, 2026 for reduced group C^* -algebras and the ball-covering property found the source paper and nearby operator-space BCP literature, but no exact reduced-group theorem. The subgroup conditional expectation is standard; the BCP application is the agent-produced partial result recorded here.

Human Review Focus

Reviewers should check:

1. Lemma 1, especially the containment of a single reduced group algebra element in a countable subgroup algebra;
2. Lemma 2, the canonical expectation $C_r^*(G) \rightarrow C_r^*(H)$;
3. the final BCP escape inequality.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Nathaniel P. Brown and Narutaka Ozawa, *C^* -Algebras and Finite-Dimensional Approximations*, American Mathematical Society, 2008.

- [3] Run `fa_banach_001`, line 19, *The Separable C^* -Algebra Case of BCP*, solution packet 2311.14257_separable_cstar_bcp_classification, 2026.